Customer Shopping Behavior Analysis: Project Report

1. Introduction and Objective

Retail businesses need to understand who their customers are, what they buy and what drives spending. This project analyses a dataset of 3,900 customer purchases to answer practical business questions about revenue, discounts, subscriptions, shipping and customer loyalty.

Objectives

- Clean and prepare the raw data for analysis.
- Answer ten business questions with SQL.
- Build an interactive dashboard so non-technical users can explore the results.

2. Dataset

Item	Detail
File	`data/customer_shopping_behavior.csv`
Rows	3,900 (one purchase per unique customer)
Columns	18 (19 after cleaning and feature engineering)
Age range	18 to 70
Purchase amount	$20 to $100
Review rating	2.5 to 5.0
Categories	Clothing, Accessories, Footwear, Outerwear (25 items, 50 US states)

3. Tools and Technologies

Python (pandas, NumPy, Matplotlib, Seaborn), SQLAlchemy and PyMySQL, MySQL, Power BI, Jupyter Notebook, Git and GitHub.

4. Methodology

4.1 Data cleaning

- **Inspection**: `shape`, `info`, `describe` showed 3,900 rows and no duplicates.

- **Missing values**: only `Review Rating` had gaps (37 rows). They were filled with the **median rating of the same category**, which keeps the fill realistic and is robust to outliers.
- **Column names**: converted to lowercase snake_case; `Purchase Amount (USD)` became `purchase_amount`.
- **Redundant column**: `Promo Code Used` was identical to `Discount Applied` on every row, so it was dropped.
- **Validation**: after cleaning, 0 missing values, 0 duplicates and 3,900 unique customer IDs.

4.2 Feature engineering

- **`age_group`**: customers split into four equal-sized quartile groups: Young Adult (18-31), Adult (32-44), Middle-aged (45-57), Senior (58-70).
- **`purchase_frequency_days`**: purchase frequency text (Weekly, Monthly, Quarterly and so on) converted to a number of days for easier comparison.

4.3 Database

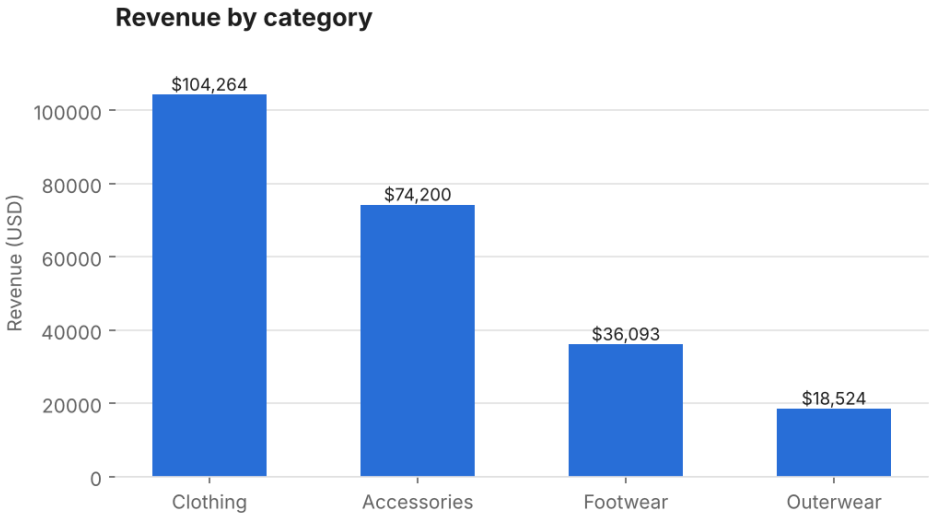
The cleaned DataFrame was loaded into a MySQL table `customer` in the database `customer_behavior` using SQLAlchemy, then queried with SQL.

4.4 Dashboard

A Power BI report with KPI cards (number of customers, average purchase amount, average review rating), column and bar charts (sales and revenue by category, sales and revenue by age group), a donut chart of subscription status, and slicers for category, gender, shipping type and subscription status.

5. Overall Results

Metric	Value
Total revenue	$233,081
Average purchase amount	$59.76
Average review rating	3.75
Customers who subscribe	1,053 (27%)
Purchases with a discount	1,677 (43%)



Category	Revenue	Share	Orders	Average order
Clothing	$104,264	44.7%	1,737	$60.03
Accessories	$74,200	31.8%	1,240	$59.84
Footwear	$36,093	15.5%	599	$60.26
Outerwear	$18,524	7.9%	324	$57.17

6. SQL Analysis: Questions and Findings

Q1. Revenue by gender. Male customers generated $157,890 and female customers $75,191. Males are 2,652 of 3,900 customers (68%), so the gap comes from customer count, not higher spending: the average purchase is $59.54 for males and $60.25 for females.

Q2. Discount users who still spent above average. 839 of the 1,677 customers who used a discount (50%) spent at least the overall average of $59.76, so discounts do not stop customers from spending.

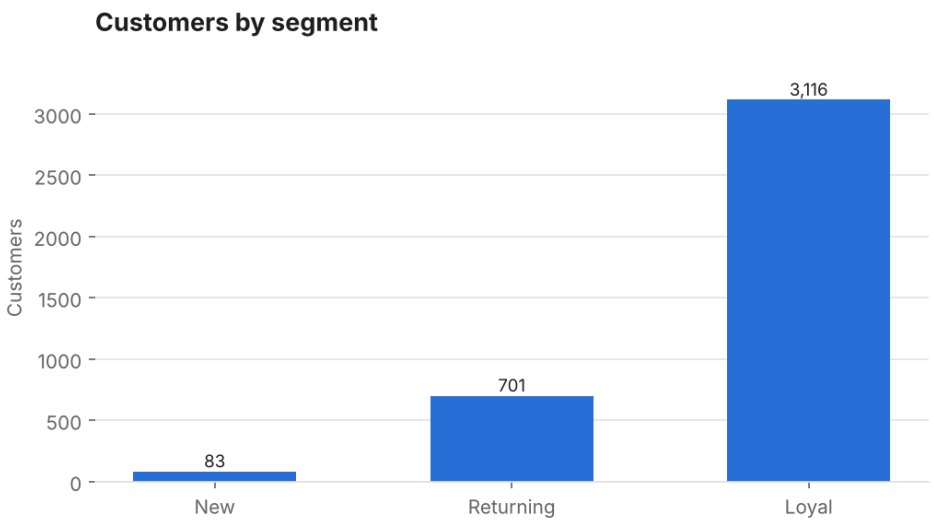
Q3. Top 5 products by average rating. Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Handbag (3.78). Differences are small, so ratings are fairly uniform across products.

Q4. Standard vs Express shipping. Express averages $60.48 and Standard $58.46, a difference of about $2.

Q5. Do subscribers spend more? No. Subscribers: 1,053 customers, $59.49 average, $62,645 total. Non-subscribers: 2,847 customers, $59.87 average, $170,436 total. Subscription does not currently translate into higher spend per purchase.

Q6. Products with the highest discount rate. Hat (50.00%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), Pants (47.37%).

Q7. Customer segments by previous purchases. New (1 previous purchase): 83. Returning (2-10): 701. Loyal (more than 10): 3,116, which is 80% of all customers.



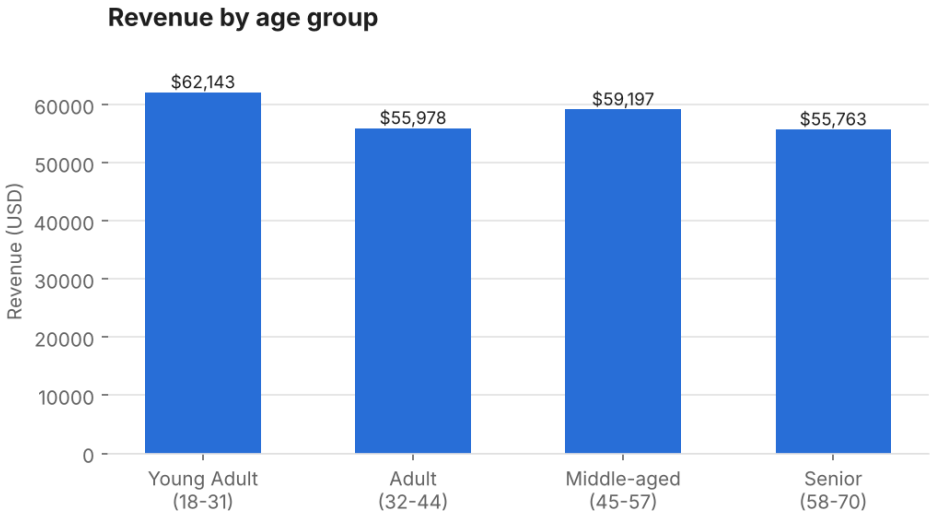
Q8. Top 3 products per category.

Category	#1	#2	#3
Accessories	Jewelry (171)	Belt (161)	Sunglasses (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	n/a

Accessories and Outerwear ties are broken arbitrarily by `ROW_NUMBER()`. Outerwear only has two products in the data.

Q9. Are repeat buyers likely to subscribe? Among customers with more than 5 previous purchases (3,476), 958 (28%) subscribe and 2,518 (72%) do not. Being a repeat buyer does not make a customer much more likely to subscribe.

Q10. Revenue by age group. Young Adult $62,143, Middle-aged $59,197, Adult $55,978, Senior $55,763. The youngest group leads, but all four groups are within about 11% of each other.



7. Key Insights

1. Clothing and Accessories bring in about 77% of revenue.
2. Revenue differences by gender, age and shipping type are mostly about how many customers are in each group, not how much each spends. Average spend is close to $60 almost everywhere.
3. Subscription does not lift spend per purchase, and only 27% of customers subscribe.
4. The customer base is very loyal (80% with more than 10 previous purchases) but there are few new customers (2%).
5. Discounts are widely used (43% of purchases) with no visible effect on average spend ($59.28 with discount vs $60.13 without).

8. Recommendations

- **Grow the subscriber base** by adding real subscriber benefits (early access, free shipping), since subscription alone is not raising spend.
- **Review discount strategy**: discounts do not increase basket size, so target them to specific high-discount products (Hat, Sneakers, Coat) or to win-back campaigns.
- **Acquire new customers**: the base is loyal but almost no one is new.
- **Focus marketing on female customers**, who are only 32% of the base but spend slightly more per purchase.
- **Promote Express shipping**, which goes with a slightly higher average purchase.

9. Limitations

- Each customer appears once, so there is no time dimension and no true repeat-purchase or trend analysis.
- Differences between groups are small and no statistical significance tests were run, so they should be read as indicative.
- Missing ratings were imputed with category medians, which slightly narrows the rating spread.
- The age-group bins are quartiles of this dataset, not standard marketing age bands.

10. Conclusion

The analysis shows a loyal, evenly spending customer base where revenue is driven by volume in Clothing and Accessories rather than by differences in spend per purchase. The cleaned data, SQL queries and Power BI dashboard give a reusable foundation; the next steps are adding date information, testing differences statistically and building a predictive model for subscription or churn.